q,t-Catalan Combinatorics

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Introduction

The Catalan Numbers are a sequence of numbers that appear in various combinatorial problems–over 200! The sequence starts as follows: $1, 1, 2, 5, 14, 42, \ldots$

Theorem 1

The Catalan numbers satisfy the recurrence relation

$$C_0 = 1 \text{ and } C_n = \sum_{i=1}^{n} C_{i-1}C_{n-i} \text{ for } n > 0$$

One object counted by the Catalan numbers are Dyck Paths– which are lattice paths that stay above the diagonal. This is the representation that we are mainly focused on.

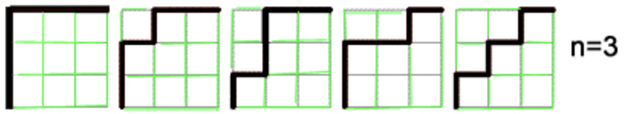


Image source: `https://media.geeksforgeeks.org/wp-content/uploads/Dyck.jpg`

Theorem 2

$$C_n = \frac{1}{n+1}\binom{2n}{n}$$

The Catalan numbers can be generalized to yield the q,t Catalan polynomials. To do this, two statistics, area and bounce, are defined on each Dyck path.

Definition. *The* **area** *of a Dyck path is the total number of complete grid squares beneath the path and above the diagonal.*

To calculate the bounce of a Dyck path, first you must draw its corresponding bounce path. The bounce path of a Dyck path of size n is a new Dyck path. To draw it, start at $(0, 0)$ and go up until the initial Dyck path is reached. Then go right until the diagonal, and repeat the process until reaching (n, n). Now the bounce can be found.

Definition. *Let* $(v_0, v_0), (v_1, v_1), \ldots, (v_m, v_m)$ *be the points at which the bounce path intersects the diagonal, where* $v_0 = 0$ *and* $v_m = n$*. The* **bounce** is equal to $\sum_{i=0}^{m-1} i(v_{i+1} - v_i)$.

Theorem 3

Let X_n be the set of Dyck paths of size n.

$$C_n(q,t) = \sum_{D \in X_n} q^{\text{area}(D)} t^{\text{bounce}(D)}$$

Remark 1. *Note* $C_n(1,1) = C_n$*. It turns out that* $C_n(q,t)$ *is symmetric, meaning that* $C_n(q,t) = C_n(t,q)$*. However, there is currently no direct combinatorial proof of this.*

Symmetry & the q,t-Catalan numbers

Definition. *A function* $f(x_1, x_2, \ldots, x_n, y_1, y_2, \ldots, y_n)$ *is symmetric in two variables if swapping any* $(x_i, y_i), (x_j, y_j)$ *yields the same polynomial.* f *is alternating in two variables if swapping any* $(x_i, y_i), (x_j, y_j)$ *yields* $-f$*.*

Definition. *A polynomial* $f(x_1, x_2, \ldots, x_n, y_1, y_2, \ldots, y_n)$ *is* **bihomogenous** with bidegree $(a.b)$ if every monomial is of the form $x_i^a y_j^b$

Fix some $n \in \mathbf{N}$ and consider the subspace V of alternating polynomials of bidegree (a, b). Let $W \subset V$ be the subspace of *basic* alternating polynomials, alternating polynomials which aren't the linear combination of alternating polynomials divisible by a nonconstant symmetric polynomial. Then the dimension of V/W is the coefficient of the $q^a t^b$ monomial in $C_n(q,t)$.

Example 1. *Consider* $C_2(q,t) = q + t$*. A basis for* V *is* $\{x_1 - x_2\}$*, which is not a linear combination of alternating polynomials divisible by a nonconstant symmetric polynomial. Therefore* W *has dimension* 0*, and* V/W *has dimension 1, corresponding to the coefficient of 1 of the monomial* $q = q^1 t^0$*.*

Knowing this, it is clear that $C_n(q,t)$ is symmetric, as if $\{f_1(x_1, \ldots, x_n, y_1, \ldots, y_n), \ldots, f_m(x_1, \ldots, x_n, y_1, \ldots, y_n)\}$ is a basis of $V^n_{(a,b)}/W^n_{(a,b)}$, then $\{f_1(y_1, \ldots, y_n, x_1, \ldots, x_n), \ldots, f_m(y_1, \ldots, y_n, x_1, \ldots, x_n)\}$ is a basis of $V^n_{(b,a)}/W^n_{(b,a)}$.

Finding Bases of Subspaces

Given a distinct set of pairs $(s_1, t_1), \ldots, (s_n, t_n)$ such that $\sum_{i=1}^{n} s_i = a$ and $\sum_{i=1}^{n} t_i = b$, define

$$\Delta_{(s_1,t_1),\cdots,(s_n,t_n)} = \begin{vmatrix} x_1^{s_1}y_1^{t_1} & \ldots & x_1^{s_n}y_1^{t_n} \\ \vdots & \ddots & \vdots \\ x_n^{s_1}y_n^{t_1} & \ldots & x_n^{s_n}y_n^{t_n} \end{vmatrix}$$

Given a (not necessarily distinct) set of pairs $(u_1, v_1), \ldots, (u_n, v_n)$ such that $\sum_{i=1}^{n} u_i = a$ and $\sum_{i=1}^{n} v_i = b$, define

$$m_{(u_1,v_1),\ldots,(u_n,v_n)} = \sum_{\sigma \in S_n} x_1^{u_{\sigma(1)}} y_1^{v_{\sigma(1)}} \cdot \ldots \cdot x_n^{u_{\sigma(n)}} y_n^{v_{\sigma(n)}}$$

It turns out that $\left\{\Delta_{(s_1,t_1),\ldots,(s_n,t_n)} : \sum_{i=1}^{n} s_i = a,\ \sum_{i=1}^{n} t_i = b, \text{ and } i \neq j \implies (s_i, t_i) \neq (s_j, t_j)\right\}$ is a basis of $V^n_{(a,b)}$, and $\left\{m_{(u_1,v_1),\ldots,(u_n,v_n)} : \sum_{i=1}^{n} u_i = a,\ \sum_{i=1}^{n} v_i = b\right\}$ is a basis for the subspace of bihomogeneous symmetric polynomials of bidegree (a, b). From this, we can construct a basis β of $V^n_{(a,b)}/W^n_{(a,b)}$, the end goal being to associate elements of β with Dyck paths of size n, area a, and bounce b.

Methods and Results

Code 1: Calculate $m \times \Delta$

An important step in the process of constructing β is, given appropriate sets of pairs $(s_1, t_1), \ldots, (s_n, t_n)$ and $(u_1, v_1), \ldots, (u_n, v_n)$, calculating the corresponding polynomials m, Δ, and $m \times \Delta$. We can do so using code. m is found by generating and summing over the unique permutations of the input exponent tuples. Δ is found by populating a matrix with monomials and finding its determinant. We have this code in C++ and Mathematica, the latter being better for our purposes.
Once we have m and Δ for any given tuples, finding $m \times \Delta$ is trivial.

Example 2. *m-Exponents:* $((2,1),(0,3))$ *and Δ-Exponents:* $((1,1),(2,2))$
Polynomial m: $x_2^2 y_2 y_1^3 + x_1^3 y_2^3 y_1$
Polynomial Δ: $x_1 x_2^2 y_1 y_2^2 - x_1^2 x_2 y_1^2 y_2$
Polynomial $m \times \Delta$: $-x_1^2 x_2^3 y_2^2 y_1^5 + x_1 x_2^4 y_2^3 y_1^4 - x_1^4 x_2 y_2^4 y_1^3 + x_1^3 x_2^2 y_2^5 y_1^2$

Example 3. *M-Exponents:* $((1,0),(1,0),(0,2))$ *and Delta-Exponents:* $((1,0),(1,0),(0,2))$
Polynomial M: $x_2 x_3 y_1^2 + x_1 x_3 y_2^2 + x_1 x_2 y_3^2$
Polynomial Delta: 0
Polynomial MxDelta: 0

Remark 2. *Notice Δ is 0 when there are non-unique tuples. This is why we can ignore them.*

Code 2: Calcluate q,t-catalan polynomial using symmetric and alternating polynomials

Program 1 defines a helper function that, given $a, b \geq 0$, generates a list of all distinct (or not distinct) n-tuples $(a_1, a_2, \ldots, a_n)$ and $(b_1, b_2, \ldots, b_n)$ such that:

$$a_1 + a_2 + \cdots + a_n = a, \quad b_1 + b_2 + \cdots + b_n = b, \quad \text{with } a_i, b_i \geq 0.$$

Program 2 computes $m \times \Delta$, written in terms of Δ's. We take all permutations of m, add it to the corresponding Δ, and, if no duplicate points are found, we add it to a dictionary of possible solutions given by the first part. This is a shortcut method to calculate $m \times \Delta$ as described in Code 1.

Program 3 takes inputs (a, b) and n, and returns the q,t-Catalan coefficient $C_n(a, b)$. We loop over all $(s_1, t_1), \ldots, (s_n, t_n)$, distinct, and $(u_1, v_1), \ldots, (u_n, v_n)$, not necessarily distinct, such that:

$$(s_1 + \cdots + s_n + u_1 + \cdots + u_n,\ t_1 + \cdots + t_n + v_1 + \cdots + v_n) = (a, b).$$

Then, we call program 2 for each of these combinations to find $m_{(u_1,v_1),\ldots,(u_n,v_n)} \cdot \Delta_{(s_1,t_1),\ldots,(s_n,t_n)}$ as a linear combination of basis elements of $V^n_{(a,b)}$, given by Program 1.
Then we can express $m_{(u_1,v_1),\ldots,(u_n,v_n)} \cdot \Delta_{(s_1,t_1),\ldots,(s_n,t_n)}$ as a vector (e.g., $[2, 3, 4, 0, 0]$). Finally, we create a matrix with these vectors as rows and our basis for $V^n_{(a,b)}$ as columns, and find the rank of the matrix. The rank of the matrix is the dimension of $W^n_{(a,b)}$, so subtracting it from the number of columns results in the coefficient for the monomial $q^a t^b$ in $C_n(q,t)$. We then row-reduce the matrix, and the free columns form a basis for $V^n_{(a,b)}/W^n_{(a,b)}$.

Example 4. $n = 5, a = 5, b = 3$*. A basis for* $V^n_{(a,b)}/W^n_{(a,b)}$ *is*

$$\left\{\Delta_{(0,0),(0,1),(1,0),(2,0),(2,2)},\ \Delta_{(0,0),(1,0),(1,1),(1,2),(2,0)}\right\}$$

Therefore the coefficient of $q^5 t^3$ *in* $C_5(q,t)$ *is* 2*.*

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